APARNA

(BIG DATA ENGINEER)

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OBJECTIVE

Experienced Data Engineer with over **5.1 years in IT**, specializing in designing and optimizing data pipelines, ETL processes, and cloud-based storage solutions. Proficient in **AWS** and **PySpark** to deliver scalable, high-performance data solutions. Skilled in collaborating with cross-functional teams to enhance data availability and performance. Committed to leveraging my expertise to drive data-driven decisions and improve operational efficiency. Seeking to contribute to impactful data engineering projects in a growth-focused organization.

PROFESSIONAL SUMMARY

- Having **5.1 years** of overall IT experience.
- Relevant around **5+ Years** of IT experience in dealing with Bigdata components like **HDFS, HIVE, SQOOP, AIRFLOW, SQL, SPARK, PYTHON,PYSPARK,AWS,S3,REDSHIFT,LAMBDA AND GLUE.**
- Experience in requirement gathering, design, and deployment of the application using Scrum (Agile) as the development methodology.
- Analyze, design and build Modern data solutions using azure pass service to support visualization of data.
- Understand current production state of applications and determine the impact of new implementation on existing business process.
- Well understanding of Hadoop architecture and various components such as **HDFS**, Job Tracker, Task Tracker, **NameNode**, **DataNode**, **Yarn**.
- Experience in installation, configuration, management and deployment of **Big Data** solutions and the underlying infrastructure of **Hadoop** Cluster.
- Extensive experience in building **Spark** applications using **Python**.
- Creating **Hive** external tables using shared meta-store instead of derby with partitioning, dynamic **Partitioning** and **buckets**.
- Having experience on creating databases, tables and views in **HIVE**, **SQL**.
- Experience in **Airflow** and workflow scheduler to manage Hadoop jobs by Direct Acyclic Graph (DAG) of actions with control flows.
- Worked on **Spark, SQL** and Datasets for faster execution of Hive queries using Spark SQL Context.

- Having exclusive experience in **SPARK** and its components like Spark Core and Spark **SQL**.
- Experienced in handling large datasets using Partitions, Spark in Memory, capabilities Broadcasts, Joins, Transformations, Actions and other during ingestion process itself.
- Experienced in loading data from the different data sources like **SQL server and MySQL** into HDFS using Sqoop and load into partitioned **Hive** tables.
- Experience in designing and executing time driven and data driven Oozie workflows.
- Skilled in ensuring data quality and integrity through validation, cleansing, and transformation operations.
- Domain experience includes **Finance**, **Retail**, **Telecommunications**.
- Prepared documentation and analytic reports, delivering summarized results, analysis and conclusions to stakeholders.
- Strong debugging and problem-solving skills with excellent understanding of application.
- Have good experience in tools like Maven, Jira, Git, WinSCP, Putty and FileZilla, Eclipse, IntelliJ.

TECHNICAL SKILLS

Big Data Skills	**: HDFS, Spark, Pyspark, Python, Hive, Sqoop, Airflow, Yarn, AWS**
RDBMS	**: MYSQL, Oracle, SQL.**
Tools	**: Winscp, putty.**
IDE	**: Eclipse, Intellij.**
OS	**: Windows, Linux.**

PROFESSIONAL EXPERIENC E

Working as a **Senior Data Engineer** at **Sarlite Infotech limited** from **Sep 2020** to Till Date.

EDUCATIONAL EXPERIENCE

Completed **B.sc Computers** From **SV UNIVERSITY** IN **2020**.

PROJECT EXPERIENCE: -

Project 4:

- Project Name : TD BANK
- Client : TD BANK
- Domain : Banking
- Environment : Hadoop, Pyspark, MySQL, Python,S3,REDSHIFT,LAMBDA,GLUE,AWS
- Role : Data Engineer
- Duration :AUG 2023 TO Till Now

Project Description:

TD is the sixth largest bank in North America by branches & serves approximately 22 million customers in several locations in key financial centers around the globe.

Roles & Responsibilities:

- Design, develop, and implement ETL pipelines using AWS Glue, PySpark, and Lambda to process, clean, and transform data from multiple sources.
- Work with structured and unstructured data from systems like MySQL, Redshift, and S3.
- Design and implement data architectures that scale to handle large data sets and high-throughput workloads using Hadoop and Spark.
- Optimize ETL processes for performance and cost-effectiveness in cloud environments like AWS.
- Create data workflows that ensure timely and reliable data processing for both batch and real-time use cases.
- Build efficient and scalable data models using best practices for managing big data in platforms like Redshift and S3.
- Ensure data is properly stored, indexed, and partitioned for efficient querying and analysis.
- Automate repetitive tasks and manual data processing using Python scripts and AWS Lambda functions.
- Set up monitoring systems to track the health and performance of data pipelines, identify issues, and resolve them proactively.
- Troubleshoot and debug data-related issues, including data discrepancies and system performance bottlenecks.
- Work with AWS services like S3, Redshift, Lambda, and Glue to manage the cloud infrastructure supporting data storage and processing.
- Maintain clear documentation for data pipelines, architecture, and processes to ensure knowledge transfer and smooth team collaboration.
- Provide status reports on data pipeline performance, issues, and enhancements to project stakeholders and leadership.
- Continuously improve the design, performance, and security of data pipelines based on feedback and evolving business needs.

Project 3:

- Project Name : CRM-MCM
- Client : CRM-MCM
- Domain : Marketing
- Environment : Hadoop, Pyspark, Hive, SQOOP, MySQL ,Python Scala,AWS
- Role : Big Data Developer
- Duration : Mar 2021 TO Jul 2023
- Team Size 9

Project Description :

CRM-MCM Bigdata: CRM Bigdata is a gateway to the crm salesforce and adobe marketing campaign management systems. RESTful API calls are made from salesforce to channel and system api microservices, this in turn access HBase data storage with phoenix and respond

to salesforce with requested crm info. Adobe receives campaign information from informatica which in turn connects to Hive Datawarehouse solution resides in Hadoop to gather this info. Salesforce uses the info for crm internal and portal requirements. Adobe uses the info for torecast the sales and derive campaign strategy by visualization. As a Hadoop bigdata architect collaborate with stakeholders of applications, other vendors and client on need basis to ensure the project developers process and quality are as per the criteria of client and industry. The project is an end-to-end Hadoop solution from architectural design, development, implementation in prod, support after implementation and train support team.

Roles & Responsibilities:

- Collaborate with stakeholders, vendors, and the client to design the architecture for the CRM-MCM project, ensuring scalability, reliability, and security.
- Develop end-to-end Hadoop-based data solutions utilizing Hadoop, Hive, HBase, and Phoenix for handling large-scale CRM and marketing campaign data.
- Design and implement efficient ETL pipelines using Apache Spark, PySpark, Scala, and Sqoop to transfer, transform, and load data from various sources (Salesforce, Adobe, Informatica) into Hadoop and Hive data warehouses.
- Develop robust data processing scripts and pipelines to extract CRM and marketing data and store it efficiently in HDFS, Hive, and HBase.
- Optimize the ETL process to handle large datasets with real-time and batch processing workflows.
- Optimize Hadoop, Spark, and Hive queries for performance to ensure low latency and high throughput when processing and analyzing data.
- Use advanced tuning techniques in HBase and Phoenix to improve data read/write efficiency and reduce bottlenecks in large data operations.
- Develop real-time data processing systems using Apache Kafka, Spark Streaming, and other big data tools to handle real-time data feeds from systems like Salesforce and Adobe.
- Create batch data processing jobs for regular extraction, transformation, and loading of marketing campaign and CRM data.
- Manage big data infrastructure on AWS cloud services, ensuring efficient deployment, resource management, and monitoring of Hadoop clusters.
- Stay updated with the latest trends in big data, cloud computing, and marketing technologies to recommend new features or optimizations for the project.

Project 2:

- Project Name : Standard Chartered Bank
- Client : Standard Chartered Bank
- Domain : Banking
- Environment : Hadoop, Pyspark, Hive, SQOOP, MySQL ,Kafka, Scala, AWS
- Role : Big Data Developer
- Duration : Oct 2019 TO Feb 2021
- Team Size 12

Project Description:

- SCB is a British conglomerate of banking and financial services operates across globe headquartered in London UK.
- SCB EDMp: Enterprise Data Management Platform is a complete end to end solution of data management for SCB's Enterprise D Management Solution to achieve one stop solution for complete enterprise data requirement.

Roles & Responsibilities :

- Design, develop, and implement end-to-end Big Data solutions using technologies such as Hadoop, Spark, Hive, and Kafka to manage large-scale banking data efficiently.
- Develop data pipelines to support SCB's Enterprise Data Management Platform (EDMp), ensuring seamless integration of financial data from multiple sources.
- Design, develop, and optimize ETL workflows using PySpark, Scala, and Sqoop to efficiently extract, transform, and load large datasets from various internal and external data sources into Hadoop-based storage systems.
- Ensure data transformation logic adheres to business requirements and maintains high performance for real-time and batch processing.
- Build scalable data architectures to integrate disparate data sources (such as MySQL, Kafka, and external APIs) into SCB's centralized data lake.
- Work with cross-functional teams to align data architecture with SCB's business objectives, ensuring data quality and security.
- Develop and implement real-time data processing solutions using Apache Kafka and Spark Streaming to ensure the efficient ingestion and processing of data in real-time for business analytics.
- Integrate Kafka streams with SCB's Enterprise Data Management Platform (EDMp) for low-latency data transfer and processing.
- Implement robust data validation and quality assurance processes within data pipelines to ensure accuracy, consistency, and integrity of financial data.
- Work with the governance team to ensure that data handling complies with SCB's data governance policies and industry standards, including data privacy regulations.
- Optimize the performance of Hadoop and Spark-based systems to ensure scalable and high-performance data processing, especially during high-load periods or batch data ingestion.
- Implement data partitioning, indexing, and caching strategies to improve query and job performance in Hive and Hadoop.
- Work with AWS cloud technologies to deploy, monitor, and manage SCB's Big Data infrastructure, ensuring that resources are optimized for cost, performance, and availability.
- Leverage AWS services like S3, Redshift, and EMR for scalable storage and computing resources.

Project 1 :

- Client : NETWORK SERVICE INTELLIGENCE
- Role : Software Engineer
- Environment : Hive, SQL, Sqoop, HDFS

- Duration : Sep 2020 to 2021 Feb
- Team Size 8

Roles & Responsibilities:

- Deployed data factory for creating data pipeline to orchestrate the data into data lake.
- Implement end-to-end ETL processes ensuring efficient data movement and transformation with a focus on data quality and governance.
- Monitoring the data pipeline
- Automating the data processing using AWS services
- Maintaining Data catalog tool
- Maintaining the data bases and users
- Providing the data solutions to the entire team
- Creating the REST APIs to retrieve analytics data from different data feeds
- Automate processes and set up monitoring for efficiency and reliability.
- Maintain comprehensive documentation for knowledge sharing and transparency.
- Building pipelines from end-to-end development on modules/functionalities independently.
- Optimize data processing workflows for performance and scalability.
- Collaborate with stakeholders including data scientists and analysts to understand and meet data requirements.